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Let the system be allowed to attain equilibrium in a cylinder with a piston. If we press the piston down to increase the pressure, the volume decreases. The system responds to this effect by reducing the number of gas molecules. i.e. it favours the formation of ammonia. If we pull the piston upwards to reduce the pressure, the volume increases. It favours the decomposition of ammonia

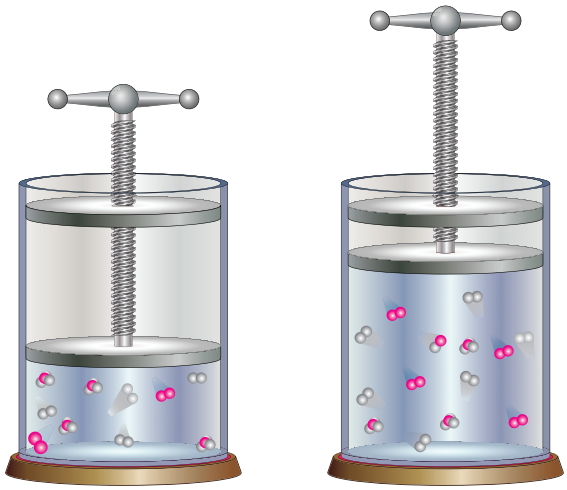


Figure 8.6 Effect of pressure on ammonia synthesis.

However, when the total number of the moles of the gaseous reactants and the gaseous products are equal, the change in pressure has no effect on system at equilibrium.

Let us consider the following reaction

$H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$

2 moles of reactants $\Rightarrow$ 2 moles of product

Here, the number of moles of reactants and products are equal. So, the pressure has no effect on such equilibrium with $\Delta n_g = 0$.

8.8. 3 Effect of temperature

If the temperature of a system at equilibrium is changed, the system responds by shifting the equilibrium in the direction that attempts to nullify the effect of temperature.

Let us consider the formation of ammonia.

$$N_2(g) + 3H_2(g) \underset{\text{endothermic}}{\overset{\text{exothermic}}{\rightleftharpoons}} 2NH_3(g) \quad \Delta H = -92.2 \text{ kJ}$$

In this equilibrium, the forward reaction is exothermic i.e. the heat is liberated while the reverse reaction is endothermic i.e. the heat is absorbed.

If the temperature of the system is increased (by supplying heat energy), the system responds by decomposing some of the ammonia molecules to nitrogen and hydrogen by absorbing the supplied heat energy. Similarly, the system responds to a drop in the temperature by forming more ammonia molecules from nitrogen and hydrogen, which releases heat energy.

We have already learnt that the change in pressure or concentration causes a change in the equilibrium concentration such that the equilibrium constant remains the same. However, in case of change in temperature, the equilibrium is reestablished with a different equilibrium constant.

8.8.4 Effect of a catalyst

Addition of a catalyst does not affect the state of the equilibrium. The catalyst increases the rate of both the forward and the reverse reactions to the same extent. Hence, it does not change the equilibrium composition of the reaction mixture.